

Michele Tosi

Quantitative Researcher | Behavioral Scientist

m.tosi.quant@gmail.com | Verona, Italy (open to relocation) | Portfolio | GitHub | LinkedIn | Google Scholar

Skills

- **Experimentation and Causal Inference:** A/B testing, experimental design, randomization, power analysis, quasi-experimental designs, mixed-effects models, hypothesis testing, statistical modeling, multivariate statistics, data analytics, descriptive and exploratory statistics, regression analysis.
- **Behavioral and UX Research:** Survey design, psychometrics, data visualization, user research, usability testing, mixed methods, behavioral segmentation, applied ML, qualitative research.
- **Programming and Machine Learning:** Python, R, SQL, Excel, Qualtrics; NLP, sentiment analysis; LLM evaluation, explainable AI, dimensionality reduction, benchmarking and model evaluation, Git.

Work Experience

Quantitative Researcher, Italian Institute of Technology, Italy Oct 2021 – Apr 2026

- Designed and analyzed randomized, controlled behavioral experiments, combining survey, behavioral, and biometric sensor data across 180+ participant sessions.
- Built end-to-end reproducible Python (pandas, NumPy, scikit-learn, statsmodels) and R pipelines for mixed-effects modeling and hypothesis testing.
- Designed quantitative evaluation frameworks for LLM outputs, combining explainable AI metrics with cognitive and behavioral modeling approaches.
- Managed end-to-end research projects, coordinating timelines, data collection, and analysis.

Quantitative Researcher, University of Oxford, UK Apr 2025 – Oct 2025

- Modeled physiological, behavioral, imaging, and clinical data to predict individual functional outcomes with statistical and machine learning approaches, under a competitively awarded fellowship.

Statistical Consultant, Freelance Sep 2023 – Sep 2024

- Advised undergraduate students and academic researchers on experimental design and psychometrics.

Survey Specialist, University of Padova and University of Verona, Italy Oct 2018 – Oct 2021

- Built survey infrastructure and performed end-to-end psychometric validation; segmented multilingual samples using clustering and dimensionality reduction for behavioral comparisons.

Selected Projects

- **Survey Optimization at Scale:** Validated a 42-item instrument on $N = 39,775$ (Cronbach's $\alpha = 0.88-0.96$) and shortened it via Random Forest feature importance while preserving predictive performance.
- **Cross-Demographic Survey Validation:** Established measurement invariance across sex and country ($N = 1,420$), enabling valid comparisons across multilingual populations.
- **LLM Evaluation and Cognitive Benchmarking:** Developed interpretable benchmarks for LLM evaluation using semantic network analysis and behavioral metrics grounded in cognitive theory.

Education

- **PhD in Brain and Cognitive Sciences**, University of Trento (CIMEC), Italy
- **MSc in Data Science**, University of Trento, Italy
- **MSc in Cognitive Psychology**, University of Padova, Italy
- **BSc in Industrial and Organizational Psychology**, University of Verona, Italy

Additional Training and Languages

Training and Grants: Data Science-driven Innovation in Business and Society (Scuola Superiore Sant'Anna, 2026); Machine Learning Crash Course (University of Genoa, 2025); Biometric Signal Processing and Behavioral Analytics (Radboud University); IIT Mobility Grant; Erasmus+; Four-year PhD Fellowship.

Languages: Italian (native), English (C2).